EPSR-test-ex8 Molecule size factor (sizefactor 2.0)

Program(s): epsr | chained passes: 4 | verification: ALL IDENTICAL (all 4 passes)

What this case tests

Exercises the `sizefactor` option of the `epsr` engine [1,2]: molecules are shrunk relative to the box at the start of a simulation to avoid entanglement, and expanded back in steps of the decline rate over successive runs. While `sizefactor` > 1 EP refinement and intramolecular atom moves are switched off [3].

Code exercised

Fortran (original, fortran_src/):

- `epsr_standalone_rev3.f`: `runepsr17` (`sizefactor` > 1: `potfac` := 0, `stepmi` := 0; decline by the rate after each run)
- `ato_manipulation_rev4.f`: `generate_mp` (WCA-truncated intramolecular core with `dcore/sizefactor`)
- `molfit.f`: `en_atom_intra` (intramolecular distances scaled by `sizefactor`), `molfit`
- `ato_read_and_save.f`: `minitc` (`dmaxmol` = `dmax·sizefactor`)

Rust (port, rust/epsr-core/src/):

- `epsr.rs`: `Epsr::setup`, `Epsr::iterate` (sizefactor decline)
- `potential.rs`: `generate_mp`; `energy.rs`: `en_atom_intra`
- `montecarlo.rs`: `molfit`; `ato.rs`: `minitc`

Physical system

The EPSR-test system: a liquid mixture of 200 dimethyl sulphoxide (DMSO, (CH3)2SO) and 400 water molecules (DMSO mole fraction 1/3; 3200 atoms) at 300 K in a cubic box of side 33.12 Å (atomic number density 0.0881 atoms/Å³). DMSO uses EPSR's Lennard-Jones + Coulomb reference potential with methyl rotational groups; water uses SPC/E Lennard-Jones and charge parameters [4] with EPSR's flexible intramolecular geometry (O-H 0.976 Å). The empirical potential is refined against three neutron diffraction data sets of deuterated DMSO in D2O, in null-scattering H2O/D2O and in H2O (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), the isotope-substitution approach of Soper and Luzar [5]. With `sizefactor` s > 1 the intramolecular bond restraints are normalised to a box s times larger (`dmaxmol` = dmax·s), so molecules are shrunk by 1/s relative to the box, and the intramolecular non-bonded core, a Weeks-Chandler-Andersen truncated n-6 potential [6], has its diameter scaled by 1/s. Here s starts at 2 and is multiplied by the decline rate 0.9 after each run whose energy is below the threshold (0).

Inputs

Copy of EPSR-test with only `sizefactor` `2.00000` `0.90000` `0.00000E+00` (factor, decline rate, threshold; factor 1.0 in EPSR-test) in `200DMSO400Water.EPSR.inp`.

How the accuracy was verified

`commands.txt` (1 program call, repeated 4 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 4 passes.
Four chained passes, each reading the reduced factor written by the previous one. The Fortran reference shows the decline 2.0 -> 1.8 -> 1.62 -> 1.458 (value in the last `.out`) and writes `sizefactor` `1.31220` to the `.inp`; as EP refinement is off, neither `.pcof` nor `delfit.dat` is written. All 25 files written are identical after every pass.

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[4] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[5] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[6] J. D. Weeks, D. Chandler and H. C. Andersen, "Role of repulsive forces in determining the equilibrium structure of simple liquids", J. Chem. Phys. 54, 5237-5247 (1971).
[7] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.